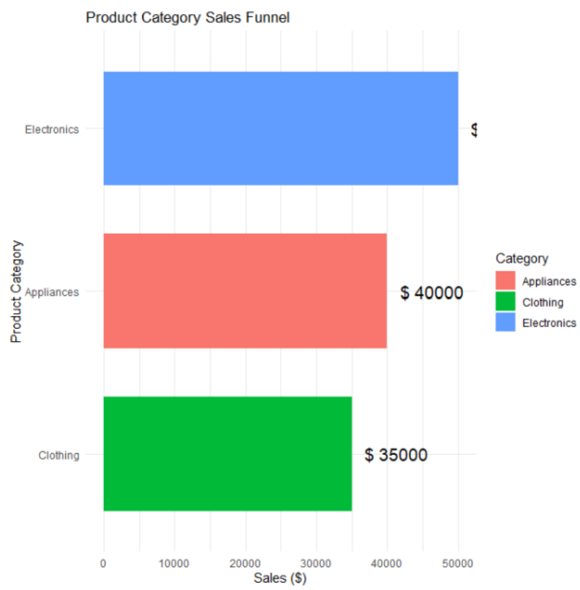
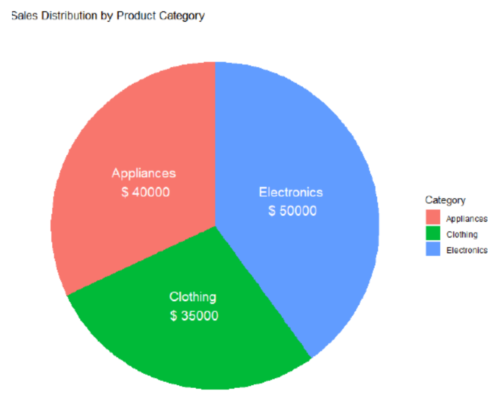
Exp 11:

```
funnel_chart <- ggplot(product_data,
                       aes(x = reorder(Category, Sales),
                           y = Sales,
                           fill = Category)) +
  geom_col(width = 0.7) +
  coord_flip() +
  geom_text(aes(label = paste("$", Sales)),
            hjust = -0.2,
            size = 5) +
  labs(
    title = "Product Category Sales Funnel",
    x = "Product Category",
    y = "Sales ($)"
  ) +
  theme_minimal()
funnel_chart
grid.newpage()
grid.table(product_data)
pie_chart <- ggplot(product_data,
                    aes(x = "",
                        y = Sales,
                        fill = Category)) +
  geom_col(width = 1) +
  coord_polar("y") +
  geom_text(aes(label = paste(Category, "\n$", Sales)),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 5) +
  labs(
    title = "Sales Distribution by Product Category"
  ) +
```

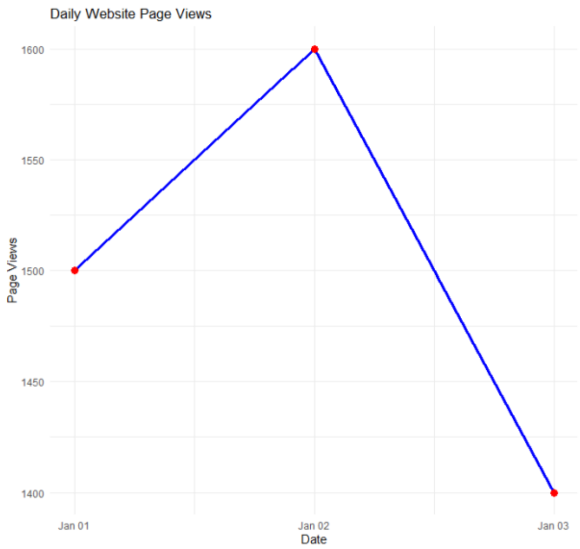




```
  theme_void()

pie_chart
```

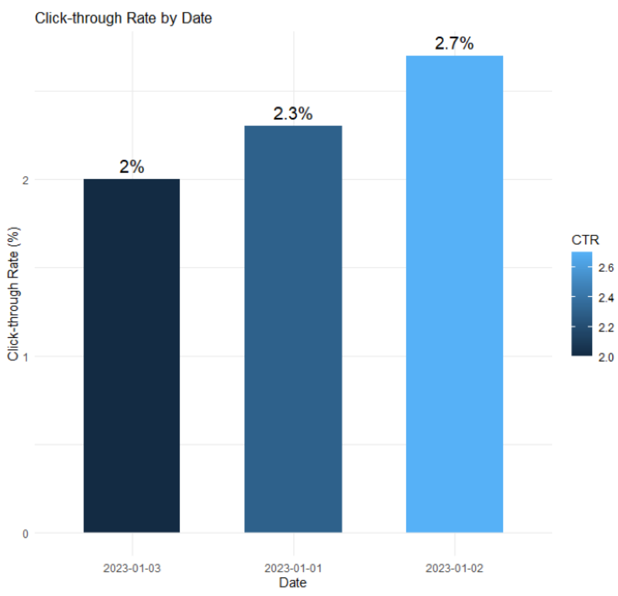
Experiment 12

```
install.packages("ggplot2")      # Run only once

install.packages("patchwork")    # Run only once

install.packages("gridExtra")    # Run only once

library(ggplot2)

library(patchwork)

library(gridExtra)

library(grid)
traffic_data <- data.frame(

  Date = as.Date(c("2023-01-01",

                  "2023-01-02",

                  "2023-01-03")),

  PageViews = c(1500,1600,1400),

  CTR = c(2.3,2.7,2.0)

)

traffic_data
line_chart <- ggplot(traffic_data,

                     aes(x = Date,

                         y = PageViews,

                         group = 1)) +

  geom_line(color = "blue", linewidth = 1.2) +

  geom_point(color = "red", size = 3) +

  labs(

    title = "Daily Website Page Views",

    x = "Date",

    y = "Page Views"

  ) +

  theme_minimal()
```




```
line_chart

bar_chart <- ggplot(traffic_data,
```

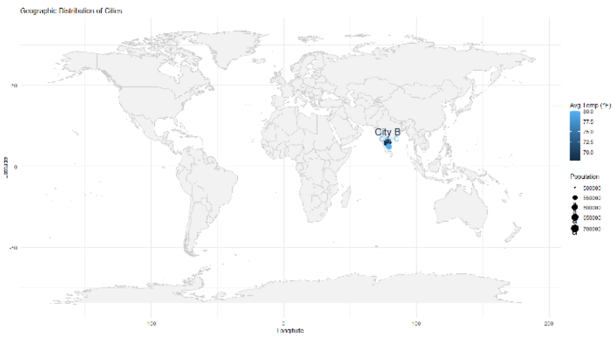
```
            aes(x = reorder(as.character(Date), CTR),
              y = CTR,
              fill = CTR)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = paste0(CTR,"%")),
        vjust = -0.5,
        size = 5) +
  labs(
   title = "Click-through Rate by Date",
   x = "Date",
   y = "Click-through Rate (%)"
  ) +
  theme_minimal()
bar_chart
```



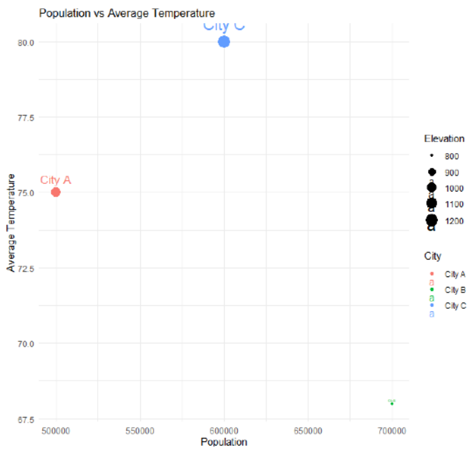
Experiment 13

```
install.packages("ggplot2")    # Run only once
install.packages("maps")       # Run only once
install.packages("patchwork")  # Run only once
install.packages("gridExtra")  # Run only once
library(ggplot2)
library(maps)
library(patchwork)
library(gridExtra)
library(grid)
geo_data <- data.frame(
  City = c("City A","City B","City C"),
  Population = c(500000,700000,600000),
  AvgTemperature = c(75,68,80),
  Elevation = c(1000,800,1200),
  Longitude = c(77.6,78.4,79.2),
  Latitude = c(13.0,14.5,12.2)
)
```

```
geo_data
map_chart <- ggplot(geo_data,
                    aes(x = Longitude,
                        y = Latitude,
                        size = Population,
                        color = AvgTemperature,
                        label = City)) +
  borders("world",
          colour = "gray80",
          fill = "gray95") +
  geom_point() +
  geom_text(vjust = -1) +
  labs(
    title = "Geographic Distribution of Cities",
    x = "Longitude",
    y = "Latitude",
    size = "Population",
    color = "Avg Temp (°F)"
  ) +
  theme_minimal()
map_chart
```




```
scatter_plot <- ggplot(geo_data,
                       aes(x = Population,
                           y = AvgTemperature,
                           color = City,
                           size = Elevation)) +
  geom_point() +
  geom_text(aes(label = City),
            vjust = -1) +
  labs(
    title = "Population vs Average Temperature",
    x = "Population",
    y = "Average Temperature",
```

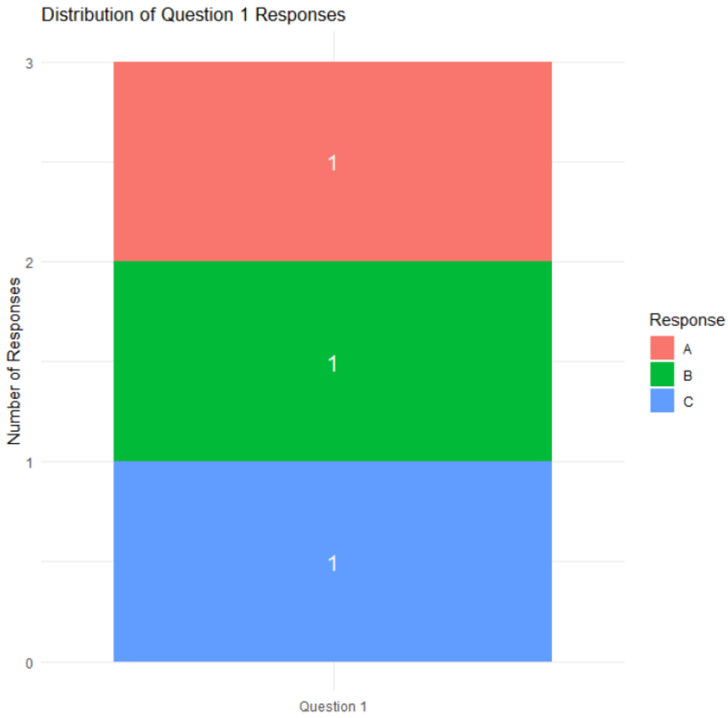


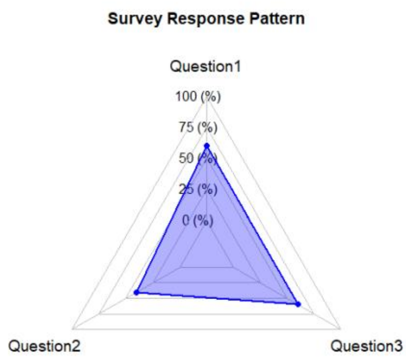
```
  size = "Elevation"
 ) +
 theme_minimal()
scatter_plot
```

Experiment 14

```
install.packages("ggplot2")     # Run only once
install.packages("tidyr")
install.packages("dplyr")
install.packages("fmsb")
install.packages("patchwork")
install.packages("gridExtra")
library(ggplot2)
library(tidyr)
library(dplyr)
library(fmsb)
library(patchwork)
library(gridExtra)
library(grid)
survey_data <- data.frame(
 Respondent = c(1,2,3),
 Question1 = c("A","B","C"),
 Question2 = c("B","A","A"),
 Question3 = c("C","D","B")
)
survey_data
q1_count <- survey_data %>%
 count(Question1)


stacked_bar <- ggplot(q1_count,
                aes(x = "Question 1",
                  y = n,
                  fill = Question1)) +
 geom_bar(stat = "identity") +
```

```
  geom_text(aes(label = n),
            position = position_stack(vjust = 0.5),
            color = "white",
            size = 5) +
  labs(
    title = "Distribution of Question 1 Responses",
    x = "",
    y = "Number of Responses",
    fill = "Response"
  ) +
  theme_minimal()
stacked_bar
radarchart(
  survey_radar,
  axistype = 1,
  pcol = "blue",
  pfcol = rgb(0, 0, 1, 0.3),
  plwd = 2,
  cglcol = "grey",
  cglty = 1,
  axislabcol = "black",
  vlcex = 1.2,
  title = "Survey Response Pattern"
)
```

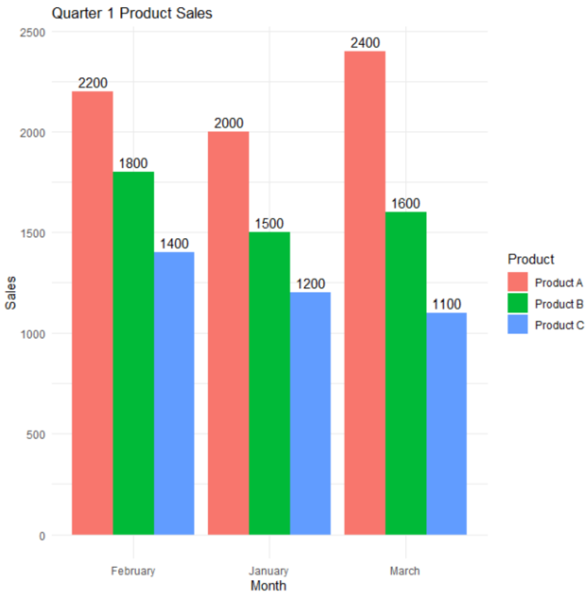




Experiment 15

```
install.packages("ggplot2")    # Run only once
install.packages("tidyr")
install.packages("dplyr")
install.packages("patchwork")
install.packages("gridExtra")

library(ggplot2)
```

```
library(tidyr)

library(dplyr)

library(patchwork)

library(gridExtra)

library(grid)
sales_data <- data.frame(

  ProductID = c(1,2,3),

  ProductName = c("Product A","Product B","Product C"),

  January = c(2000,1500,1200),

  February = c(2200,1800,1400),

  March = c(2400,1600,1100)

)

sales_data

grouped_bar <- ggplot(sales_long,

                      aes(x = Month,

                          y = Sales,

                          fill = ProductName)) +

  geom_bar(stat = "identity",

           position = "dodge") +

  geom_text(aes(label = Sales),

            position = position_dodge(width = 0.9),

            vjust = -0.4,

            size = 4) +

  labs(

    title = "Quarter 1 Product Sales",

    x = "Month",

    y = "Sales",

    fill = "Product"

  ) +

  theme_minimal()


grouped_bar
area_chart <- ggplot(sales_long,
```



```
            aes(x = Month,
              y = Sales,
              fill = ProductName,
              group = ProductName)) +
  geom_area(alpha = 0.8) +
  labs(
    title = "Overall Monthly Sales Trend",
    x = "Month",
    y = "Sales",
    fill = "Product"
  ) +
  theme_minimal()
area_chart

dashboard <- grouped_bar + area_chart +
  plot_layout(ncol = 2)

dashboard
```